P2

Practical 2: Perform ETL Process in Power BI

Aim

To perform **Extraction, Transformation, and Loading (ETL)** process to construct a database using **Power BI / SQL Server**.

BI

What is ETL?

ETL means:

E → Extract
Get data from source.

T → Transform
Clean and modify data.

L → Load
Store data in target system.

Example:

Excel file → Clean data → Load into Power BI

Step-by-Step Practical (Power BI)

Step 1: Extract Data

1. Open **Power BI Desktop**
2. Click **Home**
3. Click **Get Data**
4. Select **Excel**
5. Choose **Products.xlsx**
6. Click **Open**
7. Select **Products table**
8. Click **Edit**

Now **Power Query Editor** will open.

BI

This step is called **Data Extraction**.

Step 2: Transform Data

Now we **clean and modify the data**.

Remove Unnecessary Columns

1. Hold **Ctrl**
2. Select these columns:
    - ProductID
    - ProductName
    - QuantityPerUnit
    - UnitsInStock
3. Right click any selected column
4. Click **Remove Other Columns**

Now only important columns remain.

BI

Change Data Type

1. Select **UnitsInStock column**
2. Go to **Home**
3. Click **Data Type**
4. Select **Whole Number**

This ensures the stock values are numeric.

Create New Column (Calculation)

1. Go to **Add Column**
2. Click **Custom Column**

Write formula:

StockValue = UnitsInStock * 10

Click **OK**

Now a new column **StockValue** is created.

This is called **Data Transformation**.

Step 3: Load Data

1. Click **Home**
2. Click **Close & Load**

Now the transformed data is **loaded into Power BI model**.

BI

This step is called **Data Loading**.

Result

The data is successfully **extracted, transformed, and loaded into Power BI for analysis**.

Viva Questions (Teacher May Ask)

What is ETL?

ETL stands for **Extract, Transform, and Load**, which is used to collect data from sources, clean it, and store it for analysis.

What is Data Extraction?

It is the process of **collecting data from source systems such as Excel, SQL Server, or databases**.

What is Data Transformation?

It is the process of **cleaning, formatting, and modifying data before loading it into the target system**.

What is Data Loading?

It is the process of **storing the transformed data into a database or Power BI model**.